



Genesis Biotech Inc.

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## Anti-Denature $\beta$ -Lactoglobulin, Mouse-monoclonal Antibody

**Catalog No.** PG-20014

**Applications tested:** WB, ELISA

**Type:** IgG1

**Quantity:** 100  $\mu$ g

**Host species:** Mouse

**Clone #:** 3E7G7

**Reactivity:** bovine

**Form:** DEAE purified Ab

### Target description

Using dry milk as an immunogen, we have produced monoclonal antibodies (mAb) that can distinguish dry milk from raw milk (Chen et al., 2004). Antigenic mapping shows that the epitope is located at the residues 66-76 in strand D of thermally denatured  $\beta$ -LG (Song et al., 2005). When heated above 80 °C, the epitope region of  $\beta$ -LG is exposed and bound by the mAb (3E7G7). The antibody is therefore defined as "denatured" mAb. On the other hand, the native mAb specifically reacted with the native  $\beta$ -LG, but not heat denatured  $\beta$ -LG.

### Antigen

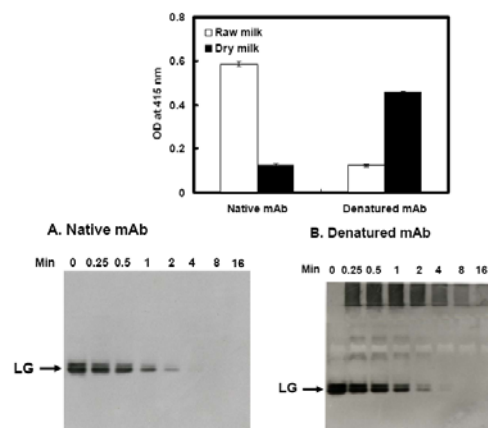
$\beta$ -lactoglobulin

### Application

ELISA, Western blotting

### Related Products

|                                             |          |
|---------------------------------------------|----------|
| Anti-native $\beta$ -LG mAb, denature mAb   | PG-20012 |
| Anti-denature $\beta$ -LG mAb, denature mAb | PG-20015 |



### Western Blot Procedure

1. Block with 1% Gelatin buffer for 1 hour at RT.
2. Wash with 0.05% PBST 3 X 10 minutes.
3. Add 3000X dilution of monoclonal antibody and Incubate for 1 hour at RT.
4. Wash with 0.05% PBST 3 X 10 minutes.
5. Add appropriate amount of correct secondary antibody, goat anti-mouse antibody conjugated with HRP). Incubate for 1 hour at RT.
6. Wash 3 X 10 minutes with 0.05% PBST at RT.
7. Add HRP substrate and develop

### ELISA Procedure

About 0.5  $\mu$ g of raw or dry milk protein in 50  $\mu$ L of PBS was coated onto a respective microtiter plate. Unbound  $\beta$ -LG was washed and subsequently blocked by an addition of 350  $\mu$ L of 1% gelatin (wt/vol) for 30 min. Following washes with PBS, 50  $\mu$ L of 3000X dilution of monoclonal antibody were added and incubated at room temperature for 60 min. Each well was then washed 3 x with gelatin-PBS containing 0.1% gelatin and 0.05% Tween-20. Bound  $\beta$ -LG antibodies were detected using a goat anti-mouse IgG conjugated with HRP in gelatin-PBS for 30 min. Finally, each well was washed and developed with 0.04% (wt/vol) 2,2-Azino-bis(3-ethylbenz-thiazoline-6-sulfonic acid) (ABTS) in PBS containing 0.01%  $H_2O_2$  (vol/vol).

### Storage

It is supplied as DEAE purified mouse monoclonal antibody (mAb) in lyophilized form. Redissolve the powder with 100  $\mu$ L sterile water will restore to the original condition. Store at 4°C for short-term application. For long-term storage, aliquot and store at -20°C.

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